IGCSE Mathematics Lesson Plan

Solving Linear Equations: Brackets, Fractions and Real-World Equations

Duration: 25 minutes | Source-based plan: supplied teacher script

Lesson overview

Topic	Solving linear equations using inverse operations, expanding brackets and clearing fractions
Core strategy	SBIC: Simplify → Balance → Isolate x → Check
Prior learning	Mental use of inverse operations; basic linear equations; multiplication and division; common multiples
Resources	Board or slides, student worksheet, pens, optional mini-whiteboards or voting response
Lesson structure	Real-world starter → explicit modelling → guided practice → misconception check → fraction example → independent practice → exit ticket

Learning objectives

- Expand brackets correctly, including when a negative number multiplies the bracket.
- Clear fractions by multiplying every term in the equation by a common multiple.
- Solve linear equations by performing the same operation on both sides.
- Check a solution by substituting it into the original equation.
- Represent and solve a simple real-world situation using a linear equation.

Success criteria

- I can expand a bracket by multiplying the outside factor by every term inside.
- I can identify a common multiple of the denominators and multiply every term by it.
- I show balanced steps and isolate x.
- I substitute my answer into the original equation and confirm both sides are equal.
- I can explain at least one step using the SBIC strategy.

25-minute teaching sequence

Time	Stage	Teacher activity	Student activity	Assessment for learning
0–3 min	Starter: Taxi challenge	Present: a taxi charges ¥10 plus ¥4 per kilometre; total fare ¥34. Ask students to solve mentally, then express it as 4x + 10 = 34.	Subtract 10, divide by 4; identify x as kilometres. Explain their mental method.	Diagnostic check: ask how they found 6 and what x represents.
3–5 min	Objectives, vocabulary and SBIC	Briefly define equation, coefficient, expand, inverse operations, LCM and substitute. Introduce SBIC and the balance/scales analogy.	Repeat or use vocabulary; state why the same operation is applied to both sides.	Cold-call: Why must an operation be applied to both sides?
5–8 min	I DO: brackets	Model 3(x + 2) = 18: expand to 3x + 6 = 18; subtract 6; divide by 3; check by substitution.	Track each step and identify the inverse operation used.	Ask what 3 multiplies when expanding; ask how the answer is checked.
8–11 min	WE DO: negative bracket	Guide students through 5 − 2(x + 1) = 3x − 2. Emphasise that −2 multiplies both terms inside the bracket.	Suggest each next step; simplify to 3 − 2x = 3x − 2, then solve to x = 1.	Check signs during expansion and ask students to verify both sides equal 1.
11–13 min	Misconception check	Show the error: 4(x − 3) simplified incorrectly to 4x − 3. Ask students to vote on the error.	Choose an option and explain that 4 must multiply both x and −3.	Use responses to decide whether to re-model expansion.

13–17 min	I DO / WE DO: fractions	Use the implied example (x + 3)/3 + x/6 = 4. Identify LCM 6; multiply every term by 6 to get 2(x + 3) + x = 24; expand and solve.	Identify LCM; follow the transformation to 3x + 6 = 24 and x = 6.	Ask why the right-hand side must also be multiplied by 6.
17–21 min	Independent practice	Assign worksheet questions. Students should show steps and use SBIC. Support students with the scaffold; ask early finishers to solve the fraction challenge.	Work independently, then check answers by substitution where possible.	Circulate/check expansion, sign handling, clearing fractions and balanced operations.
21–25 min	Real-world link and exit ticket	Briefly connect to gaming café model 2(3x + 6) = 60, then set exit ticket 3(2x − 1) + 5 = 20.	Identify what x represents in the context; solve exit ticket independently.	Collect or scan exit tickets. Expected answer: x = 3.

Worked examples to model

Example 1: Expanding brackets

3(x + 2) = 18

3x + 6 = 18

3x = 12

x = 4

Check: 3(4 + 2) = 18 ✓

Example 2: Negative factor and brackets

5 − 2(x + 1) = 3x − 2

5 − 2x − 2 = 3x − 2

3 − 2x = 3x − 2

5 = 5x

x = 1

Check: left side = 1; right side = 1 ✓

Example 3: Clearing fractions

(x + 3)/3 + x/6 = 4

Multiply every term by 6: 2(x + 3) + x = 24

2x + 6 + x = 24

3x + 6 = 24

3x = 18

x = 6

Differentiation

- Support: provide the SBIC reminder; ask students to identify the outside factor and each term in the bracket; scaffold one operation per line.
- Core: students solve the negative-bracket example and the fraction equation independently, showing all working.
- Challenge: students explain why every term must be multiplied by the LCM, then solve the harder fraction equation from the script: (2x − 1)/3 − (x + 2)/4 = 5.
- EAL support: keep the vocabulary visible, use the balance/scales analogy, and provide sentence frames such as “I subtract ___ from both sides because ___.”

Assessment for learning: evidence and response

- Starter response reveals whether students can use inverse operations and interpret a variable.

- Questioning during modelling checks expansion and equation balance.
- Misconception vote diagnoses failure to distribute a factor to every term.
- Independent practice reveals whether students can transfer SBIC to fractions.
- Exit ticket: 3(2x − 1) + 5 = 20; expected solution x = 3. If students expand incorrectly or fail to balance, revisit that step next lesson.

Teacher notes / source boundaries

- The supplied script specifies a 25-minute online demo lesson; this plan keeps that duration.
- The script names Support/Core/Challenge answers (x = 5, x = 9, x = 17) but does not include the corresponding question stems. They are therefore not reproduced as worksheet questions.
- The first fraction equation is reconstructed from the transformations explicitly described in the script: (x + 3)/3 + x/6 = 4.
- The harder fraction equation is reconstructed from the stated LCM-12 transformation: (2x − 1)/3 − (x + 2)/4 = 5.